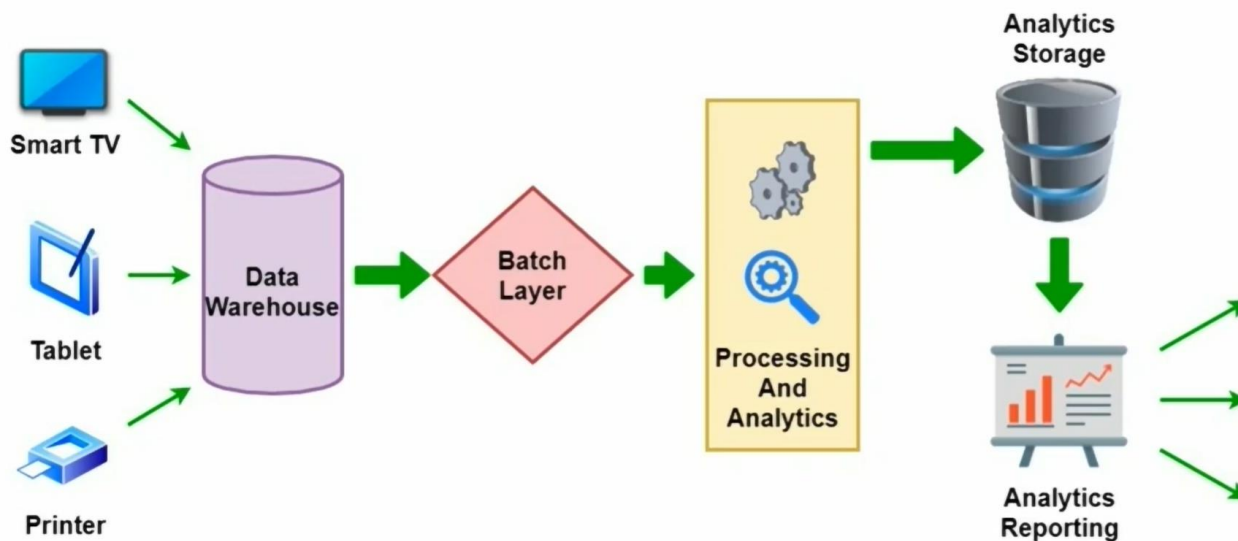


Batch Processing



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1



- Executing set of task collected over period of time
- Non – interactive and is executed in background
- Scheduled at certain time
- Some preprocessing while storing the data

Advantages

- Improves Efficiency
- Simpler than Stream Processing
- Data Quality improvement due to less user interaction

Challenges

- Troubleshooting
- High initialization costs
- Required lot of data engineering and analytics skills

Batch Processing – Architectural Components



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2

- Storage and Partitioning to manage volume and parallel processing
- Computing Requirement including Scalability
- DWH
- Fault Tolerance
- Processing Frameworks - Orchestration
- Handling Shared Data

IOT

- Mini Batch processing can be done at Edge or at Device .
- Eg aggregation, reduced data transfer to upstream, local action and local display

Batch Processing – Tools and Frameworks



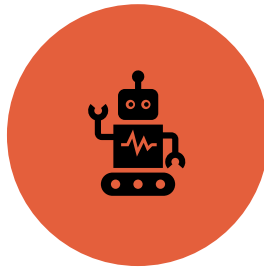
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3



STORAGE

HDFS, HBASE,
DYNAMO DB, S3,
AZURE BLOB



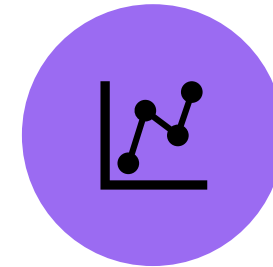
PROCESSING

HADOOP,
SPARK,
AWS EMR



ANALYTICS STORAGE DATA WAREHOUSE

AWS REDSHIFT,
DYNAMODB,
HIVE, MONGODB,
HBASE



ANALYTICS & REPORTING

TABLEAU, POWERBI,
QUICK SIGHT

Batch Processing – Spark



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4

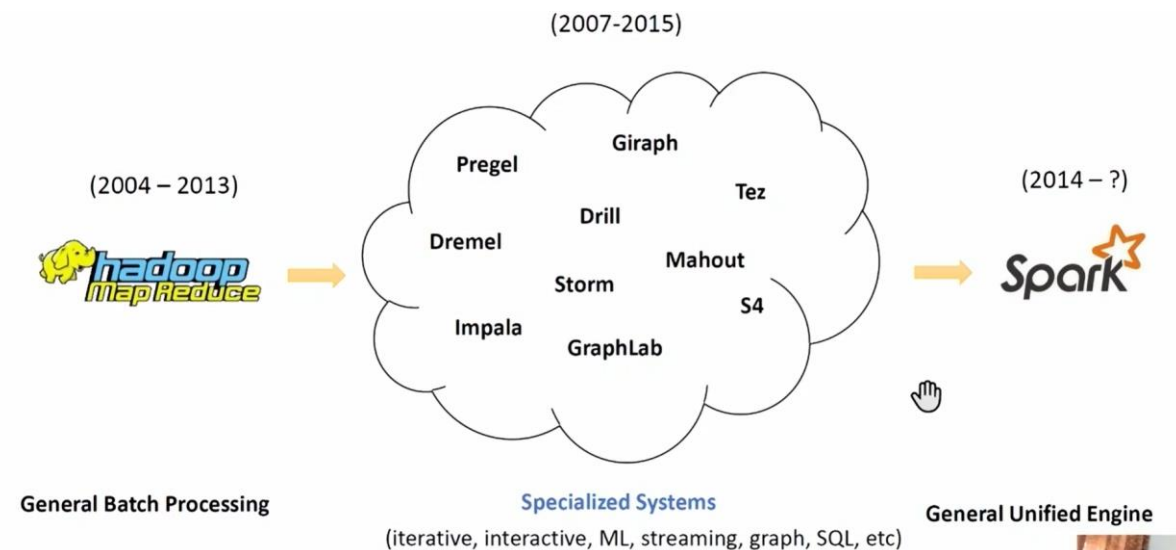
<https://spark.apache.org/> - Unifies Engine for large scale analytics

Before Spark, **MapReduce** was the Engine for analytics

- Slow & Complex
- Required data on HDFS
- Emergence of ML required lot of iterative processing

AMPLab project was taken up to find an alternative and thus emerged Spark

- Execution Engine but 100x faster
- Easy to program – Supports java , scala, python, R, SQL
- Platform Independent - Run it anywhere (works with Hadoop, Standalone cluster, cloud, K8 etc.)
- Read and write data from anywhere (RDBMS, NoSQL DB, S3, texts file)
- Spark has become Morden day ETL for Big data

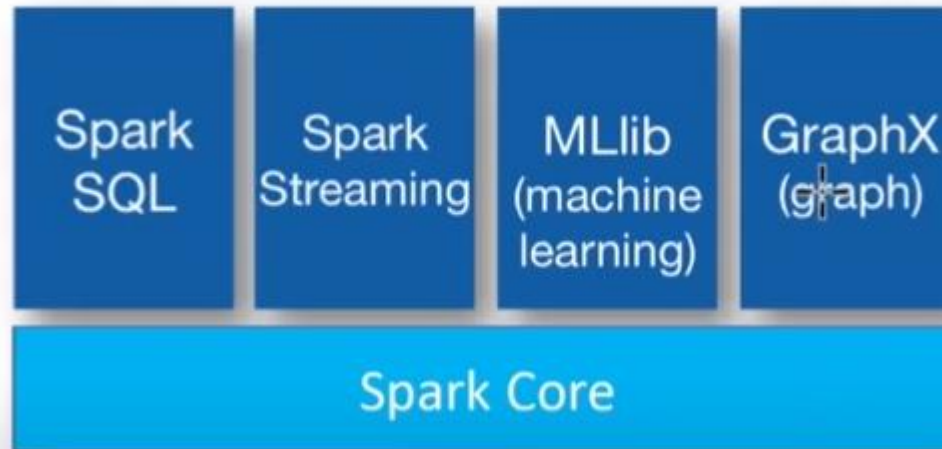


Batch Processing – Spark



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5



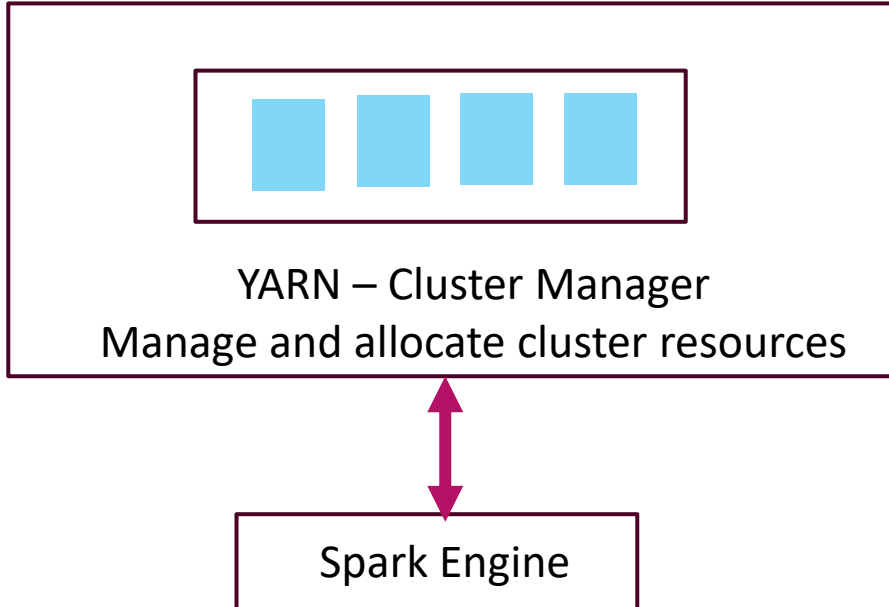
Unified Analytics Engine unlike Hadoop/MapReduce ecosystem which has so many tools

Batch Processing – Spark



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6



- When you run a spark program – it's submitted to YARN (cluster Manager)
- 2 parts of program
 - Driver – contains all configuration and starts on one of the machine. Driver talks to YARN and will negotiate with YARN for resources
 - Executors – the actual processing which can be running in parallel
- Mesos – another Cluster Manager
- Spark can also run on K8 and can also run standalone
- Spark standalone will have its own cluster manager

Hands-On



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Hands-On